Executive brief: profit-aware retention targeting

To: Head of Customer Retention | **From:** Retention analytics team | **Decision requested:** approve a 90-day retention pilot

Recommendation

Stop allocating the retention budget to "the customers most likely to leave". Allocate it to the customers for whom **churn risk x customer value x the offer's save rate** most exceeds the cost of the offer, and give each customer an offer their segment qualifies for. Start with a **randomised pilot** on the recommended list: one wave of 1,550 customers (22% of the 7,043-customer base), half held out as a control group. Scale up only if the measured save rate clears that list's **13.9% break-even**.

Why: the same budget returns more

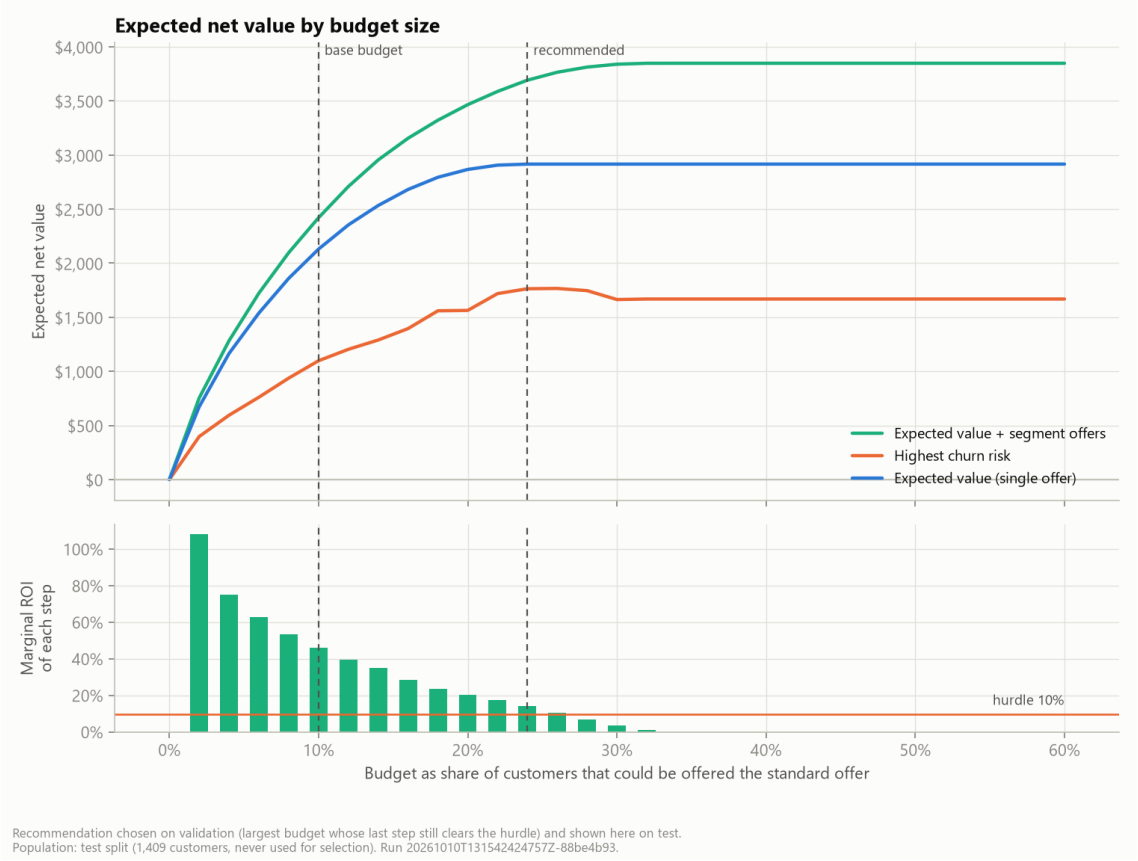
Results for 1,409 held-out customers with a fixed $3,500 budget (enough to offer 10% of them the standard offer):

Option	Customers contacted	Expected net value	ROI	Precision among contacted	High-risk/high-value coverage
Do nothing	0	$0	n/a	n/a	0.0%
Call highest churn risk	140	$1,098	31%	70.0%	3.9%
Expected value, standard offer	140	$2,130	61%	62.1%	96.1%
Expected value + segment offers	**152**	**$2,420**	**69%**	**59.9%**	**100.0%**

- **Better targeting earns $1,032 more (94%) for the same spend.** Using what these customers actually did, the gain is $1,308 (95% interval $509 to $2,153).
- **Why:** risk-only targeting spends the budget on high-risk customers whose remaining value is too small to repay an offer. It reaches only 4% of the high-risk, high-value customers, against 100% for the recommended rule. It has higher precision but less value.
- **Segment offers add $290** under the assumed save rates. Checked against actual outcomes the gain is not distinguishable from zero (-$27), so the pilot must test the offers, not just the list.

How much to spend

Each extra budget step returns less (figure). We keep only the steps that still earn at least a 10% marginal return; this was decided on validation data. That rule recommends **$6.00 per customer in the base** (enough for the standard offer to 24% of customers; about $42,238 for all 7,043). On the test customers this buys 310 offers (22% of customers) for $8,450, an expected net value of $3,692 (ROI 44%). Beyond 27% of customers contacted, nobody left has positive expected value, so further budget should not be spent.



How sure are we?

- **The model ranks customers well:** ROC-AUC 0.83, PR-AUC 0.62 (random contact: 0.27). Predicted churn is within 2.0 percentage points of observed churn on average. It predicts who leaves, not who an offer would keep.
- **Break-even:** the base-budget list stays profitable while the true save rate is above **11.8%**, the larger recommended list above **13.9%**; we assume 20%. Below 8.9% nobody is worth contacting.
- **Scenarios:** expected net value is $0 pessimistic (no campaign is run), $2,420 base and $18,525 optimistic. A list frozen under the pessimistic case would lose $2,797.
- **Biggest uncertainties:** save rate of the standard offer and value horizon (months).

Risks and guardrails

- **No causal evidence yet.** Save rates are documented assumptions. Contacting customers can even raise churn, so the pilot keeps a random control group.
- **Sample data.** The IBM sample describes a fictional US company with 26.5% churn; real US carriers lose about 1% of postpaid customers per month. The results show that the method works, not how big the prize is. Retrain on our own data before rollout.
- **Fairness and experience.** Senior customers are contacted 3.7x as often as others, because they are higher-risk and pay more; age and gender are not model inputs. Loyal, low-risk customers rarely get offers, so watch for "loyalty penalty" complaints.
- **Compliance (Vietnam).** Prior consent and a Do-Not-Call check are required. Advertising calls are limited to one per 24 hours, between 08:00 and 17:00 (Decree 91/2020). A data-protection impact assessment is required under the Personal Data Protection Law 2025.

Decisions requested

1. Approve one pilot wave: 775 customers contacted and 775 held out. Offer and contact budget about $21,125.
2. Name owners: Retention (offers and script), Call centre (capacity), Finance (margin and offer cost), Legal/DPO (consent and DPIA).
3. Agree the decision rule now. After 90 days: **go** (scale to the recommended budget) if the offer effect is statistically significant and the measured save rate is at least 13.9%; **stop** if it is clearly below; otherwise run one more wave. If the true save rate is 20%, this rule says go 94% of the time.